

Neural Networks Questions

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Question 1: Which of the following are the reasons behind the transition to deep learning?

- a. Machine learning algorithms are out-dated
- b. *More computation power*
- c. Deep learning uses neural networks which are always better than machine learning algorithms
- d. *More data*

Question 2: What is the main reason behind using activation functions?

- a. To add dependencies
- b. *To break linearity*
- c. To add linearity
- d. To break dependencies

Question 3: Which of the following represents a batch of size 32 of 16 pixels by 16 pixels grayscale images?

- a. *(32, 1, 16, 16)*
- b. (1, 32, 16, 16)
- c. (1, 16, 32, 16)
- d. (1, 16, 16, 32)

Question 4: What does a neuron compute in neural networks?

- a. A neuron computes a linear function ($z = Wx + b$)
- b. A neuron computes an activation function followed by a linear function ($z = Wx + b$)
- c. *A neuron computes a linear function ($z = Wx + b$) followed by an activation function*
- d. A neuron computes a linear function ($z = Wx + b$) followed by averaging the result

Question 5: Consider the following variables a and b, with a.shape = (256, 64) and b.shape = (64, 128). What is the shape of a @ b, np.dot(a, b) and a * b?

- a. a @ b = (256, 128), np.dot(a, b) = ERROR, a * b = ERROR
- b. a @ b = (256, 128), np.dot(a, b) = (256, 128), a * b = ERROR
- c. a @ b = (256, 128), np.dot(a, b) = (256, 128), a * b = (256, 128)
- d. a @ b = ERROR, np.dot(a, b) = (256, 128), a * b = ERROR

Question 6: Suppose the value of z is a negative number. What will be the output of the relu activation function? i.e what will be the value of $relu(z)$?

- a. The absolute value of z
- b. The same value of z
- c. *Zero*
- d. Error as relu activation function does not take negative input

Question 7: The following are hyperparameters EXCEPT:

- a. Epochs
- b. Hidden Units
- c. Learning Rate
- d. *Loss Function*

Question 8: Assume a neural network architecture with 2 nodes in the inputs layer, 3 nodes in the hidden layer and 2 nodes in the output layer. What is the total number of learnable parameters in this network?

- a. 12
- b. 10
- c. *17*
- d. 16

Question 9: Assume a neural network architecture with 784 nodes in the inputs layer, 16 nodes in 2 hidden layers and 10 nodes in the output layer. What is the total number of learnable parameters in this network?

- a. 12990
- b. 13000
- c. 12988
- d. *13002*

Question 10: Why is Pytorch library the go to library for deep learning?

- a. *Support of dynamic computational graphs*
- b. *Flexible utilizaiton of GPUs*
- c. Scalibility
- d. Stability\$\$